

SQL Fundamentals - SELECT and Filtering

01) Syntax: SELECT *

FROM employees;

Outcome:

employees

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Maidoo	IT	72 000	2017-09-30	Johannesburg

02) SELECT department,

FROM employees

WHERE department <> 'IT';

Outcome:

department
HR
Finance
Marketing

03) Syntax: SELECT first_name, last_name, salary

FROM employees

WHERE salary ORDER BY salary DESC;

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FROM employees

ORDER BY salary DESC;

Outcome:

employees

first_name	last_name	salary
Eva	Naidoo	72 000
Alice	Green	70 000
Cathy	Zulu	65 000
David	Mokoena	50 000
Brian	Lee	45 000

04) Syntax : SELECT id, first_name, last_name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;

Outcome

id	first_name	last_name	salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

02) Filtering with WHERE

05) Syntax : SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'IT';

Outcome

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

06) `SELECT id, first_name, last_name, department, salary`
`FROM employees`
`WHERE department = 'Finance' AND salary > 60000;`

Outcome

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65000

07) Syntax : `SELECT id, first_name, last_name, department`
`FROM employees`
`WHERE department = 'HR'`
`OR department = 'Marketing';`

Outcome

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

08) Syntax : `SELECT id, first_name, last_name, department`
`FROM employees`
`WHERE NOT department = 'IT';`

Outcome

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

09) SELECT id, first_name, last_name, department
 FROM employees
 WHERE department IN ('HR', 'IT', 'Finance');

Outcome

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10) Syntax : SELECT *

FROM employees

WHERE department = 'IT'

AND WHERE salary > '65 000'

AND WHERE city = 'Johannesburg';

Outcome

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72 000	Johannesburg